LESSON 01 · CHEAT SHEET

Numbers, Text & Formatting

VARIABLES & TYPES

```
reward = 4860            # int    whole number
days = 12.5              # float  decimal
name = "Wren Hawkridge"  # str    text
is_open = True           # bool   True / False
leader = None            # "no value" (VBA: Nothing)
type(days)               <class 'float'>
```

ARITHMETIC

`17 / 7`	2.428...	always a float
`17 // 7`	2	whole part (INT)
`17 % 7`	3	remainder (MOD)
`2 ** 3`	8	power (Excel: 2^3)
`round(2.718, 2)`	2.72	ROUND
`0.1 + 0.2`	0.30000000000000004	normal, round at the end

TEXT (STRINGS)

`"Wren" + " " + "Hawkridge"`	join (VBA: `&`)
`" hi ".strip()`	"hi" (TRIM)
`"wREN".title()`	"Wren" (PROPER)
`"Hi".upper()` / `.lower()`	UPPER / LOWER
`"a-b".replace("-", "")`	"ab" (SUBSTITUTE)
`len("Wren")`	4 (LEN)
`"Q-42".find("-")`	1 (FIND minus 1)
`"-" * 5`	"-----" (REPT)

Chain them: `" q-1 ".strip().upper()` → `"Q-1"`

SLICING: LEFT, RIGHT, MID

```
text = "GOB-0042"
        01234567    ← positions start at 0
text[0]       "G"       # first character
text[-1]      "2"       # last character
text[:3]      "GOB"     # LEFT(text, 3)
text[-4:]     "0042"    # RIGHT(text, 4)
text[4:6]     "00"      # MID(text, 5, 2)
```

`text[a:b]` runs from a up to, **not including**, b. It has b−a characters.

CONVERTING

```
int("42")          42
float("12.5")      12.5
str(42)            "42"
float("1,500 cr".replace(" cr", "").replace(",", ""))
                   1500.0
" Yes ".strip().lower() == "yes"     True
```

F-STRINGS: NUMBER FORMATS

```
name = "Wren"; share = 2708.333
f"Hi {name}!"          Hi Wren!
```

`{share:.2f}`	2708.33	2 decimals
`{share:,.2f}`	2,708.33	+ thousands
`{4860:,}`	4,860	thousands
`{0.125:.1%}`	12.5%	percent
`{42:04d}`	0042	zero-padded
`{"ab":<6}`	"ab "	left in 6
`{"ab":>6}`	" ab"	right in 6

TRUE OR FALSE

```
level == 19            # equal (VBA: =)
level != 19            # not equal (VBA: <>)
400 <= reward <= 1200  # between, limits included
party >= 3 and difficulty <= 4
not is_open
```

VBA → PYTHON

`Dim x As Long`	`x = 5`
`' comment`	`# comment`
`Debug.Print x`	`print(x)`
`CInt / CDbl / CStr`	`int() / float() / str()`
`Format(x, "#,##0.00")`	`f"{x:,.2f}"`
`And / Or / Not`	`and / or / not`

GOTCHAS

- `^` is not power. Use `**`
- `=` stores, `==` compares
- `Reward` ≠ `reward` (case-sensitive)
- `"5" + 3` is an error. Convert first
- `round(2.5)` is 2: halves round to even, like VBA's `Round`